

Yuanhang Gao(高源航)

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Personal Information

- PhD candidate in NUDT, supervised by Prof. Xiangrui Yang and Prof. Dongsheng Li. Received the Master’s degree in cyberspace security from NUDT, in Jun 2025.
- Interest: LLM Inference, System for AI.

Education

National University of Defense Technology	✉ Changsha	PhD	2026.3 ~ now
National University of Defense Technology	✉ Changsha	Master	2022.9 ~ 2025.6
Zhengzhou University	✉ Zhengzhou	Bachelor	2018.9 ~ 2022.7

Research & Competitions

Magneto: Load-Balanced Key-Value Service for Write-Intensive Workloads 2023.10 ~ 2024.7
Research Project, Project Leader

- Designed a write-buffering module on the data plane of in-network switches to address load imbalance caused by write-intensive workloads. By delaying write operations and merging multiple writes to the same key, the system effectively balances backend server loads. Developed a fault-tolerance mechanism to ensure service availability and data recoverability under extreme conditions.
- Outcomes: 1 x TSC (CCF-A), first author; 1 x ISPA (CCF-C), first author.

CAInNet: A In-Network Computing Platform for AI Acceleration 2023.6 ~ 2023.8
Future Network Technology Innovation Competition

- Designed a platform based on FPGA-implemented PISA (Protocol-Independent Switch Architecture) to reduce data movement and enable pipelined processing of data and packets along the transmission path. Refactored the PISA architecture to support SIMD and MIMD modes, integrated AI computing units and parameter configuration modules. Each pipeline stage performs one layer of neural network computation; combined stages complete inference.

Honors & Awards

- Excellent Student Scholarship, NUDT – Oct 2023
- Future Network Technology Innovation Competition – National Grand Prize – Aug 2023
- Outstanding New Student Scholarship, NUDT – Oct 2022
- "Three Good Student" Award, Zhengzhou University – Dec 2019